Topics for exam MPA-MLR

Area A, Ing. Vičar

1) Optimization in machine learning.
2) Evaluation metrics and class imbalance.
3) Validation approaches and preprocessing.
4) Feature selection.
5) Feature reduction.
6) Linear models in machine learning.
7) Linear and kernel regression.
8) Bayes rule and its use in machine learning, basics of probability.
9) Maximum likelihood estimation.
10) Maximum a-posteriori probability and Bayesian inference.
11) Naive Bayes classifier.
12) Gaussian mixture model and Logistic regression.

Area B, Ing. Chmelík

1) Bias-Variance Tradeoff.

2) “Bagging” – Bootstrap Aggregation and “AdaBoost” – Adaptive Boosting.

3) Decision trees and Random Forest.

4) Artificial Neural Networks (shallow, fully connected) – principle, basic blocks, function, differences with deep NN.

5) ANN optimization algorithms.

6) Regularization techniques for ANNs (DropOut, BatchNorm, Weight Decay, basic approaches).

7) Convolutional Neural Networks – principle, basic blocks, function.

8) Special blocks in CNN - 1x1 convolution, concatenation, dilated convolution, skip connection, residual block, bottleneck.

9) Up-sampling methods in CNN (unpooling, interpolation, transposed convolution).

10) Task dependent CNN architectures and their applications.

11) Transformers, self-attention, tokenization and embeddings.